5/8/26

MLA0403- Deep Learning

Roll no: 25

"CO2- AT3"

P. Bhanu Prakash
192425263

—Class Test

① Difference between Machine Learning & Deep Learning.

Machine Learning	Deep Learning
uses simple algorithm	uses neural networks Algorithm
It needs less data	It consume larger data.
Features are selected manually	Features are learned automatically.
Faster to tain	Takes more training time

Applications:

→ Machine Learning - spam emails detection, House price prediction

→ Deep learning - Face recognition, self-driving cars

② Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.

Ans: Representation learning

→ It is the Process where a neural network automatically learns useful features from data.

Width

→ More neurons in one layer

→ Learns more information but needs more memory

Depth

→ More hidden layers

→ learns complex patterns but takes more time to train

③ Compare the ~~cor~~ ReLU, Leaky ReLU (LReLU), and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behaviour, advantages, disadvantages and suitable use cases.

Ans:

ReLU

→ Output is 0 for negative values

→ Fast and simple

→ May cause dead neurons.

Leaky ReLU

→ Gives a simple small values for negative inputs

→ Reduces dead neurons problems

→ ~~P~~ Slightly slower than ReLU

ELU

→ Gives smooth negative values

→ Better learning performance

→ More computation

Uses:-

* ReLU : CNNS

* Leaky ReLU : Deep neural networks

* ELU : Image recognition

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(4) Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzman Machine (RBMs) and Autoencoders and compare their objectes and applications

Ans

Unsupervised learning:

→ Unsupervised learning is learns patterns from data without labels.

→ It means unlabeled data provided in unsupervised leaning.

Restricted Boltzman Machine (RBM's)

→ Learns hidden features from input data.

→ Used in recommendation systems.

Autoencoders

→ Compresses data and reconstructs it.

→ Used for image compression and noise removal.

Difference:

→ RBM is a probabilistic Model

→ Autoencoder is a neural network for data encoding and decoding.

5) Explain the concept of Autoencoders and it's uses in Deep Learning.

Ans:-

Autoencoder

An Autoencoder is a neural network that learns to compress input data and reconstruct it. It works without a labeled data, so it is an unsupervised learning technique.

Uses:

→ Image Compression

→ Noise removal

→ Feature extraction

→ Anomaly detection

→ Dimensionality reduction.